

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
Department of Computer Science and Engineering
ASSESSMENT TOOL 2 – Short Answer

Course Code:	ITA03	Course Title:	Mobile computing
CO Assessed:	CO1	Assessment:	Assessment Tool 2 – Short Answer
Weightage:	30 % of CIA	Total Marks:	100 Marks
CO1 Statement:	Analyze the mobile network components, mobile base station, mobile infrastructures and protocols in cellular systems.		
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Section / Batch:	CSE-DS (Slot A)	Date:	24/07/2026

Code of Conduct

I, B. Niketha (192373034) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: Niketha B

Instructions

- This test contains 50 questions, Total: 100 Marks.
- Duration: 1 hour.

100%

Short Answers

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Course code: ITA03

Given Amount of spectrum = 45 MHz

$$\text{Full duplex} = 2 \times \text{Simplex} = 2 \times 25 \\ = 50 \text{ kHz}$$

$$\text{Total channels} = \frac{45 \times 10^6}{50 \times 10^3} = \frac{45}{50} \times 10^3 = 900$$

$$\boxed{\text{Total channels} = 900}$$

② Given Total channels = 560

$$\text{cluster size} = 4$$

$$\text{channels per cell} = \text{Total channels} \div \text{cluster size}$$

$$560 \div 4$$

$$\boxed{= 140}$$

③ Given Carriers = 100

$$\text{Time slots per carrier} = 8$$

$$\text{Total voice channels} = \text{carriers} \times \text{Time slots}$$

$$= 100 \times 8$$

$$\boxed{= 800}$$

Given

$$\text{Base station power} = 20 \text{ W}$$

$$\text{Handset power} = 4 \text{ W}$$

$$\text{Power ratio} = 10 \log_{10} (P_1/P_2)$$

$$= 10 \log_{10} (20/4)$$

$$= 10 \log_{10} (20)$$

$$= 10 \times 1.301$$

$$\boxed{13 \text{ dB}}$$

Given

$$\text{Carrier Signal} = -60 \text{ dBm}$$

$$\text{Interference} = -92 \text{ dBm}$$

$$c/i = \text{Carrier} - \text{Interference}$$

$$= (-60) - (-92)$$

$$\boxed{= 32 \text{ dB}}$$

⑥ Spectrum = 42 MHz
 channel bandwidth = 300 kHz
 cluster size = 4
 Total channels = spectrum ÷ channel bandwidth
 channels per cell = total channels ÷ cluster size
 (Total channels = $42,000 \div 300 = 140$)
 (channels per cell = $140 \div 4 = 35$)

⑦ Given
 Initial received power = 64 W
 Distance doubled
 Path-loss exponent = 4
 New power = old power ÷ 2^4

$64 \div 16$
 $= 4$ W

⑧ Given
 cell radius doubled
 Inverse-square law ($n=2$)
 Required power = $(2^2) \times$ original power
 $4 \times$ original power
 Transmitted power must increase by 4 times

⑨ cluster size = 4
 channels per cell = 15
 Total channels = cluster size $\times$ channels per cell

4×15
 $= 60$ channels

⑩ Given
 Speed = 100 km/h
 frequency = 900 MHz
 converted speed:
 $100 \times 1000 / 3600 = 27.78$ m/s

$$f_d = \frac{v \times f}{c}$$

where $v = 3 \times 10^8 \text{ m/s}$
 $f = 900 \times 10^6 \text{ Hz}$
 $c = 3 \times 10^8 \text{ m/s}$

$$f_d = \frac{3 \times 10^8 \times 900 \times 10^6}{3 \times 10^8}$$

$$f_d = 83 \text{ Hz}$$

Frame duration = 5ms

No. of slots = 8

$$\text{slot duration} = \frac{\text{Frame Duration}}{\text{No. of slots}}$$

$$\left[\frac{5}{8} = 0.625 \text{ ms} \right]$$

Total bandwidth = 24 MHz = 24,000 kHz

Carrier spacing = 200 kHz

$$\text{No. of carriers} = \frac{\text{Total Bandwidth}}{\text{Carrier spacing}}$$

$$\left[\frac{24000}{200} = 120 \right]$$

Given

Transmit power = 8 W

Path loss = 33 dB

Convert watts to dBm

$$P(\text{dBm}) = 10 \log_{10}(P \times 100)$$

Received Power = $P_r = P_t - \text{Path loss}$

$$\text{Transmit power} = 10 \log_{10}(8000) = 39 \text{ dBm}$$

$$\text{Received power} = 39 - 33 = 6 \text{ dBm}$$

Given

Frequency = 1900 MHz

Distance = 2.5 km

$$\text{FSPL} = 32.44 + 20 \log_{10}(d) + 20 \log_{10}(f)$$

$$= 32.44 + 20 \log_{10}(2.5) + 20 \log_{10}(1900)$$

$$32.44 + 4.96 + 65.58$$

$$= 105.98 \approx 106 \text{ dB}$$

(15) Given

No. of cells = 6

Calls per cell = 250

Total Capacity = cells $\times$ calls per cell

$$6 \times 250 = 1500$$

(16) Calls per second = 4000

Average call duration = 3 min = 180 sec

Effective calls = calls/sec $\times$ call duration

$$4000 \times 180 = 720000$$

(17) Given

channels = 150

Bandwidth per channel = 20 kHz

total Bandwidth = channels $\times$ Bandwidth per channel

$$150 \times 20 = 3000 \text{ kHz}$$

$$= 3.0 \text{ MHz}$$

(18) Guard band = 40 kHz

No. of guard bands = 29

total Guard Band = $(30-1) \times 40$

$$29 \times 40 = 1160 \text{ kHz}$$

$$= 1.16 \text{ MHz}$$

(19) Given

speed = 90 km/h

cell radius = 1.5 km

cell diameter = 3 km

Handoffs = $\frac{\text{Dist Travelled}}{\text{Cell Diameter}}$

Dist travelled in 1 hour = 90 km

$$= \frac{90}{3} = 30$$

Given

Regional centers = 8

Base stations per center = 40

users per base station = 250

total users = Centers $\times$ Base station $\times$ Users per Base station

$$8 \times 40 \times 250$$

$$= 80000$$

Given

$$\text{Total area} = 400 \text{ km}^2$$

$$\text{Cell radius} = 400 \text{ m} = 0.4 \text{ km}$$

$$\text{Cell area} = R^2$$

$$\text{No. of cells} = \frac{\text{Total Area}}{\text{Cell Area}}$$

$$= \frac{400}{0.16} = 2500 \text{ cells}$$

$$= 2500 \text{ cells}$$

22) Total bandwidth = 6 MHz = 6000 kHz

$$\text{Channel bandwidth} = 30 \text{ kHz}$$

$$\text{No. of channels} = \frac{\text{Total Bandwidth}}{\text{Channel Bandwidth}}$$

$$= \frac{6000}{30} = 200$$

23) Subscribers = 1200

$$\text{Cell per hour} = 4$$

$$\text{Cell duration} = 2.5 \text{ min}$$

$$\text{Traffic} = \frac{\text{Subscribers} \times \text{calls/hr} \times \text{call Duration}}{60}$$

$$= \frac{1200 \times 4 \times 2.5}{60}$$

$$= \frac{12000}{60} = 200$$

Given

$$\text{Total bandwidth} = 40 \text{ MHz} = 40,000 \text{ kHz}$$

$$\text{Bandwidth per user} = 400 \text{ kHz}$$

$$\text{users} = \frac{\text{Total Bandwidth}}{\text{Bandwidth per user}}$$

$$= \frac{40000}{400} = 100$$

25

Base stations = 1200

users per station = 600

Total users = stations $\times$ users per station

$$= 1200 \times 600 = 720000$$

26

total cells = 200

Cells per paging round = 4

Paging Rounds = $\frac{\text{total cells}}{\text{cell per Round}}$

$$= \frac{200}{4} = 50$$

27

Capacity = 100,000

call attempts = 40,000/min

call duration = 3 min

Active offered traffic

= Call Attempts $\times$ Duration

Blocked fraction = $\frac{\text{offered traffic} - \text{Capacity}}{\text{offered traffic}} \times 100$

$$= 40000 \times 3 = 120000$$

$$= \frac{120000 - 100000}{120000} \times 100$$

$$= \frac{20000}{120000} \times 100$$

$$= 16.6\%$$

28

Subscribers = 600,000

Roaming = 15%

Roaming users = $\frac{15}{100} \times 600000$

$$= 90000$$

29

Given

Capacity = 12,000

Current load = 9,000 calls/s

$$\text{utilization} = \frac{\text{current load}}{\text{Capacity}} \times 100$$

$$= \frac{9000}{13000} \times 100$$

$$= 69.23\%$$

30) Total spectrum = 48 MHz

$$= 48,000 \text{ kHz}$$

$$\text{channel bandwidth} = 200 \text{ kHz}$$

$$\text{Reuse factor} = 4$$

$$\text{Total channels} = \frac{\text{spectrum}}{\text{channel bandwidth}}$$

$$\text{channels per cell} = \frac{\text{Total channels}}{\text{Reuse factor}}$$

$$= \frac{2400}{200} = 12$$

$$= \frac{240}{4} = 60$$

31) Total bandwidth = 30 MHz

$$= 30,000 \text{ kHz}$$

$$\text{channel spacing} = 200 \text{ kHz}$$

$$\text{No. of channels} = \frac{\text{total Bandwidth}}{\text{channel spacing}}$$

$$= \frac{30000}{200} = 150$$

32) Handoff attempts = 2500

$$\text{Success rate} = 99\%$$

$$\text{Successful Handoffs} = \frac{99}{100} \times 2500$$

$$= 0.99 \times 2500 = 2475$$

33) Transmitt power = 12 W

$$\text{Path loss} = \text{und B}$$

$$\text{Convert watts to dBm:}$$

$$P(\text{dBm}) = 10 \log_{10} (P \times 1000)$$

$$\text{Received power} = P_r = P_t - \text{path loss}$$

$$10 \log_{10}(12000) = 40.8 \text{ dBm}$$

$$= 40.8 - 42 = -1.2 \text{ dBm}$$

$$\boxed{= -1.2 \text{ dBm}}$$

34

No. of cells = 7

users per cell = 900

Total capacity = cells $\times$ users per cell

$$7 \times 900 = 6300$$

$$\boxed{= 6300}$$

35

Frame duration

$$= 4.8 \text{ ms}$$

No. of slots = 8

$$\text{slot Duration} = \frac{\text{Frame Duration}}{\text{no. of slots}}$$

$$= \frac{4.8}{8} \boxed{= 0.6 \text{ ms}}$$

36

Speed = 150 km/h

frequency = 1.8 GHz = 1800 MHz

$$\frac{150 \times 1000}{3600} = 41.67 \text{ m/s}$$

$$u = 41.67 \text{ m/s}$$

$$\therefore f_d = \frac{uf}{c}$$

$$f = 1.8 \times 10^9 \text{ Hz}$$

$$c = 3 \times 10^8 \text{ m/s}$$

$$f_d = \frac{41.67 \times 1.8 \times 10^9}{3 \times 10^8}$$

$$\boxed{f_d = 250 \text{ Hz}}$$

37

Capacity = 250,000 calls

Call rate = 70,000 calls/min

Average duration = 4 min

offered traffic = call rate $\times$ Duration

$$\text{Blocked percentage} = \frac{\text{offered} - \text{capacity}}{\text{offered}} \times 100$$

$$= \frac{70000 \times 4 - 250000}{70000 \times 4} \times 100$$

$$= \frac{280000 - 270000}{280000} \times 100$$

$$= \frac{10000}{280000} \times 100$$

$$\therefore \boxed{= 10.71\%}$$

Cells = 9

users per cell = 600

Total users = cells $\times$ users per cell

$$\therefore \boxed{9 \times 600 = 5400}$$

total calls = 60,000 calls/s

Drop rate = 0.4%

$$\text{Dropped calls} = \frac{0.4}{100} \times 60000$$

$$\therefore \boxed{= 0.004 \times 60000 = 240}$$

Given

Transmit power = 43 dBm

Received power = -87 dBm

Path loss = $P_t - P_r$

$$= 43 - (-87)$$

$$43 + 87 = \boxed{130 \text{ dB}}$$

41

Area per station = 2.75 km^2

Stations = 150

Total area per station $\times$ No. of stations

$$2.75 \times 150 = \boxed{375 \text{ km}^2}$$

42

Spectrum = 48 MHz = 48,000 kHz

channel = 200 kHz

Reuse factor = 6

channels per cell = (Spectrum $\div$ channel Bandwidth) $\div$ Reuse factor

$$(48000 \div 200) \div 6$$

$$\boxed{\frac{240}{6} = 40}$$

43

Given

$$\text{paging cycles} = \text{total users} \div \text{users per cycle}$$

$$= 150000 \div 6000$$

$$\therefore T = 25$$

44

Given

$$\text{Handoffs} = \text{call duration} \div \text{handoff interval}$$

$$\therefore H = \frac{9}{3} = 3$$

45

Given

$$\text{SNR} = \text{Signal} - \text{Noise}$$

$$= (-8\text{ dB}) - (-11\text{ dB})$$

$$\therefore \boxed{= 3\text{ dB}}$$

46

Given

$$\text{Bandwidth} = \text{users} \times \text{Bandwidth per user}$$

$$= 250 \times 150$$

$$= 37500\text{ kbps} = \boxed{37.5\text{ Mbps}}$$

47

Given

$$\text{traffic} = (\text{subscribers} \times \text{call} \times \text{duration}) \div 60$$

$$(1500 \times 4 \times 2.5) \div 60$$

$$\frac{15000}{60} = \boxed{250}$$

48

Given

$$\text{total Bandwidth} = \text{channels} \times \text{BW per channel}$$

$$250 \times 12$$

$$= 3000\text{ kHz}$$

$$\boxed{= 3\text{ MHz}}$$

49

Given

$$\text{Throughput} = \text{connections} \times \text{Data Rate}$$

$$= 1500 \times 600$$

$$= 900,000\text{ kbps}$$

$$\boxed{= 900\text{ Mbps}}$$

Given

$$\text{Radius} = 4\text{ km}$$

$$\text{Area} = \pi R^2$$

$$\text{No. of towers} = \text{total area} \div \text{tower coverage area}$$

$$= 3.14 \times 4^2$$

$$= 50.24\text{ km}^2$$

$$\text{No. of towers} = \frac{350}{50.24} = \boxed{7}$$